

417521 - MACHINE LEARNING

IN-SEM EXAM PREPARATION NOTES — UNIT I

Introduction to Machine Learning

Programme: Fourth Year Artificial Intelligence and Data Science (AIDS)

University / Pattern: Savitribai Phule Pune University — 2020 Course

Subject: 417521 Machine Learning In-Sem: 30 Marks

Unit I: 06 Hours Course Outcome: CO1 — Describe and compare different models of Machine Learning.

SYLLABUS COVERAGE

- Introduction: What is Machine Learning, Definitions, Real-life applications, ML vs Traditional Programming, ML vs AI vs Data Science.
- Learning Paradigms: Learning Tasks — Descriptive and Predictive Tasks; Supervised, Unsupervised, Semi-supervised and Reinforcement Learning.
- Models of Machine Learning: Geometric, Probabilistic, Logical, Grouping and Grading, Parametric and Non-parametric Models.
- Feature Transformation: Dimensionality Reduction techniques — PCA and LDA.

Prepared as detailed, easy-to-understand Unit I notes strictly aligned with the syllabus shared for the SPPU AIDS 2020 course.

UNIT I. Introduction to Machine Learning

EXACT SYLLABUS CHECKLIST

Official Unit I point	Covered in these notes
What is Machine Learning	Yes
Definitions	Yes
Real-life applications	Yes
ML vs Traditional Programming	Yes
ML vs AI vs Data Science	Yes
Learning Tasks — Descriptive	Yes
Learning Tasks — Predictive	Yes
Supervised Learning	Yes
Unsupervised Learning	Yes
Semi-supervised Learning	Yes
Reinforcement Learning	Yes
Geometric Model	Yes
Probabilistic Models	Yes
Logical Models	Yes
Grouping Models	Yes
Grading Models	Yes
Parametric Models	Yes
Non-parametric Models	Yes
Dimensionality Reduction	Yes
PCA	Yes
LDA	Yes

This checklist is included so that no explicit Unit I point is skipped.

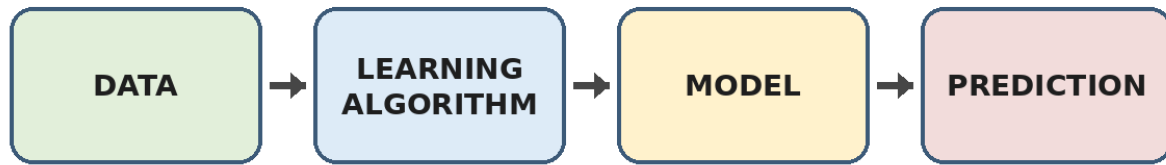
1. What is Machine Learning?

Suppose you want a computer to identify whether an email is Spam or Not Spam. In traditional programming, you may manually write rules such as “if the email contains lottery, win money or free prize, mark it as spam.” But there can be thousands of different spam messages. It is practically impossible to manually write every rule.

In Machine Learning, instead of writing every rule, we provide many examples. The learning algorithm studies those examples, learns useful patterns and then uses the learned model to make a prediction for a new email.

Simple definition: Machine Learning is a branch of Artificial Intelligence in which a computer system learns useful patterns from data or past experience and uses those patterns to make predictions or decisions.

Basic Machine Learning Process



Remember: DATA → LEARNING ALGORITHM → MODEL → PREDICTION

2. Important Definitions of Machine Learning

Definition 1 — Simple exam definition: Machine Learning is a field of study that gives computers the ability to learn from data and experience without being explicitly programmed for every task.

Definition 2 — Conceptual definition: Machine Learning builds a model from examples and uses that model to perform a task on new data.

Alpaydin's viewpoint: Machine learning involves optimizing a performance criterion using example data or past experience.

Three words to remember: DATA → LEARNING → PREDICTION

3. Important Terms in Machine Learning

Term	Meaning	Example
Dataset	Collection of data used for analysis or learning.	Student marks dataset
Feature	An input characteristic/property.	Study hours, attendance
Label / Target	The output that we want to predict.	Pass / Fail
Training Data	Data used to teach the model.	Past labelled examples
Testing Data	Data used to check performance on unseen data.	Held-out examples
Model	Learned representation/pattern used for prediction.	House-price model

4. Real-Life Applications of Machine Learning

Healthcare: Disease prediction, medical diagnosis, patient risk prediction and medical image analysis.

Banking and Finance: Fraud detection, credit-risk analysis, loan prediction and customer analysis.

E-Commerce: Product recommendation, customer segmentation, sales prediction and personalized shopping.

Spam Detection: Learning from previously classified emails and predicting whether a new email is spam.

Movie and Music Recommendation: Using previous viewing/listening behaviour, ratings and searches to recommend content.

Transportation: Traffic prediction, travel-time prediction, demand prediction and route-related prediction.

Agriculture: Crop prediction, yield prediction, disease detection and agricultural data analysis.

Customer Analysis: Studying customer behaviour and identifying useful patterns.

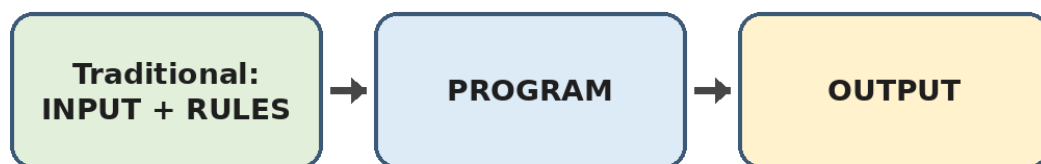
EXAM QUESTION: Explain any five real-life applications of Machine Learning. For a 6-mark answer, write the definition followed by five applications and a short explanation/example of each.

5. Machine Learning vs Traditional Programming

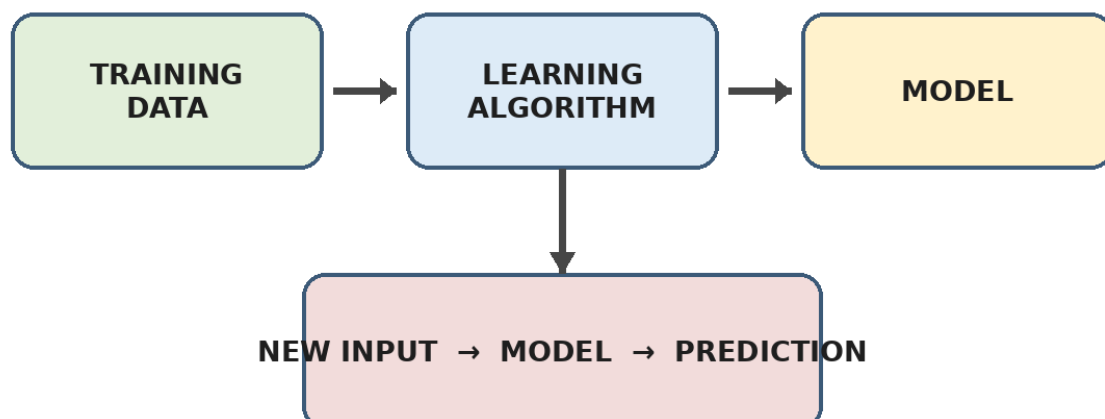
Traditional Programming: The programmer explicitly writes the rules. For example, if marks ≥ 40 , output Pass; otherwise output Fail.

Machine Learning: We provide examples and allow the algorithm to learn the relationship. Training data and an algorithm produce a model; the model is then used on new data.

Traditional Programming vs Machine Learning



Machine Learning



Traditional Programming	Machine Learning
Programmer writes rules.	Algorithm learns patterns.
Rules are explicitly specified.	Rules/patterns are learned from data.

Traditional Programming	Machine Learning
Input + program → output.	Data + learning → model; new input → prediction.
Useful when rules are known.	Useful when rules are difficult to define manually.
Program changes when rules change.	Model can be retrained with new data.
Example: Calculator.	Example: Spam detection.

MEMORY: Traditional Programming = Rules are given. Machine Learning = Rules/patterns are learned.

6. Machine Learning vs AI vs Data Science

Artificial Intelligence (AI): AI is the broader field concerned with building systems capable of performing tasks associated with intelligent behaviour, such as reasoning, decision making, problem solving, planning and learning.

Machine Learning (ML): ML is a major approach within AI in which systems learn patterns from data. Example: predicting whether a bank transaction is fraudulent.

Data Science: Data Science focuses on obtaining useful information, knowledge and insights from data. It can involve data collection, cleaning, statistics, analysis, visualization and Machine Learning.

AI	ML	Data Science
Broad field.	Major learning approach within AI.	Data-focused field.
Creates intelligent systems.	Learns patterns from data.	Extracts insights from data.
Includes reasoning, planning, learning etc.	Uses learning algorithms.	Uses statistics, programming, ML and visualization.
Example: Intelligent agent.	Example: Spam prediction.	Example: Customer-sales analysis.

MEMORY: AI = Intelligence | ML = Learning from data | Data Science = Working with data to obtain insights.

EXAM QUESTION: Differentiate between Artificial Intelligence, Machine Learning and Data Science.

7. Learning Tasks

Your syllabus contains two learning tasks: Descriptive Tasks and Predictive Tasks.

8. Descriptive Learning Task

A descriptive task tries to understand and describe patterns that already exist in the data. It mainly asks: "What patterns are present in the data?"

Example: An online shopping company studies customer data and discovers that customers aged 18-25 frequently buy headphones, customers buying laptops often buy laptop bags, and weekend purchases are higher. The system is describing existing patterns.

Definition: A descriptive learning task identifies, summarizes and describes important patterns or relationships present in existing data.

MEMORY: DESCRIPTIVE = DESCRIBE WHAT IS PRESENT

9. Predictive Learning Task

A predictive task uses known data to predict an unknown or future outcome. It asks: “What is likely to happen?”

Example: Previous house data such as area, number of rooms, location and previous prices are used to build a model that predicts the price of a new house.

Definition: A predictive learning task uses patterns learned from existing data to predict an unknown or future outcome.

Descriptive	Predictive
Describes existing patterns.	Predicts unknown/future outcomes.
Focuses on understanding.	Focuses on prediction.
Future prediction is not necessary.	Prediction is the main goal.
Example: Customer grouping.	Example: House-price prediction.

MEMORY: D = Describe | P = Predict

10. Supervised Learning

The word supervised means that the model receives guidance in the form of known answers. Supervised learning uses labelled data.

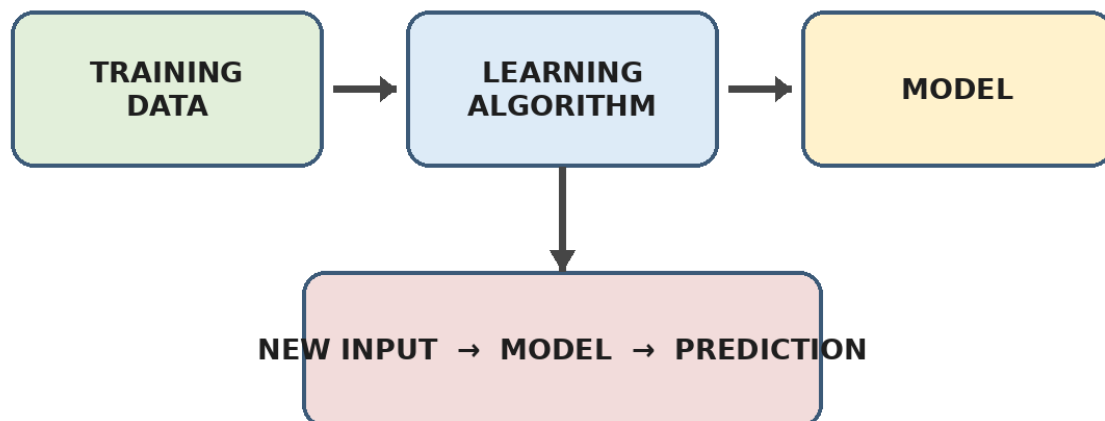
Study Hours	Result
2	Fail
4	Pass
6	Pass
8	Pass

Here, Study Hours is the input and Result is the output/label. The model learns the relationship and can predict the result for a new student.

Definition: Supervised learning is a machine learning paradigm in which a model learns from labelled training data to predict an output for new input data.

Two basic types are Classification (category output such as Spam/Not Spam or Pass/Fail) and Regression (numerical output such as house price, salary or temperature). Detailed regression is Unit II, so it is not expanded here.

Machine Learning



MEMORY: LABELLED DATA → TRAIN → MODEL → NEW DATA → PREDICTION

11. Unsupervised Learning

In unsupervised learning, there is no known output/label. The model receives data and tries to discover its structure or patterns.

Example: A company has customer information such as age, income, purchase frequency and spending amount, but no customer categories. The algorithm may discover groups such as high-spending, medium-spending and low-spending customers. This is an example of grouping/clustering.

Definition: Unsupervised learning is a machine learning paradigm in which a model learns patterns, structures or groups from unlabeled data.

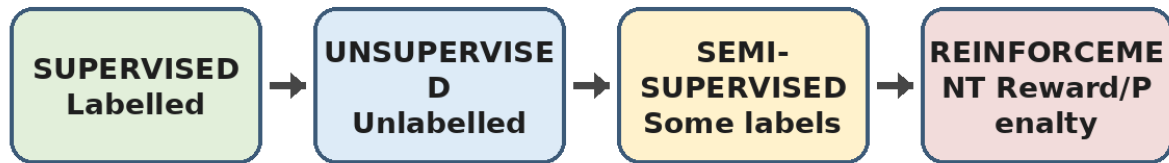
MEMORY: SUPERVISED → Answer is given | UNSUPERVISED → Answer is NOT given

12. Semi-Supervised Learning

Semi-supervised learning is a combination of supervised and unsupervised learning ideas. It uses a small amount of labelled data and a large amount of unlabelled data.

Example: Suppose there are 10,000 medical images. 1,000 are labelled and 9,000 are unlabelled. Obtaining labels from experts can be expensive and time-consuming, so a semi-supervised approach can use both types of data.

Learning Paradigms



Definition: Semi-supervised learning is a machine learning paradigm that uses a combination of labelled and unlabelled data for learning.

MEMORY: Semi = Some labelled + lots of unlabelled

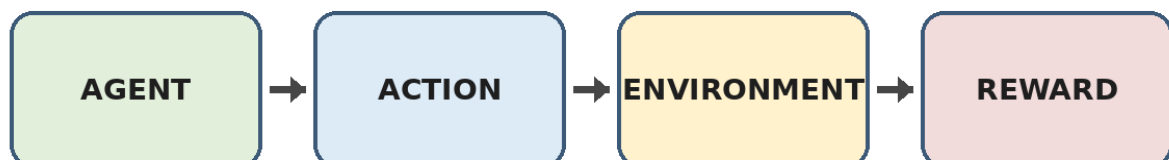
13. Reinforcement Learning

Reinforcement learning is different from supervised learning. There is no fixed correct answer for every input. Instead, an agent interacts with an environment and learns using rewards and penalties.

Example — Robot: A robot learns to reach a destination. Moving toward the destination may give a positive reward, while hitting an obstacle may give a penalty. After many attempts, it learns which actions are better.

Term	Meaning
Agent	Learner / decision maker
Environment	World in which the agent operates
State	Current situation
Action	Decision taken by the agent
Reward	Positive or negative feedback

Reinforcement Learning



Definition: Reinforcement learning is a machine learning paradigm in which an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

MEMORY: Agent → Action → Environment → Reward → Learning

14. Comparison of Learning Paradigms

Feature	Supervised	Unsupervised	Semi-Supervised	Reinforcement
Data	Labelled	Unlabelled	Labelled + unlabelled	Experience
Correct answer	Available	Not available	Available for some data	Reward-based feedback
Main goal	Prediction	Pattern discovery	Learn with limited labels	Learn best action
Feedback	Label	No direct label	Partial label	Reward/penalty
Example	Spam detection	Customer grouping	Image classification	Robot navigation

MEMORY CHART: Supervised = Teacher gives answers | Unsupervised = Find patterns yourself | Semi-supervised = Some answers + many unanswered examples | Reinforcement = Learn through reward/penalty.

15. Models of Machine Learning

Your syllabus includes Geometric, Probabilistic, Logical, Grouping and Grading, Parametric and Non-parametric models.

Models of Machine Learning



16. Geometric Model

A geometric model represents data using a geometrical representation. Data can be represented as points, lines, planes, distances or regions. Features can be represented as points in 2D/3D space and similarity can be established using lines/planes or distances.

Example: If the X-axis represents study hours and the Y-axis represents marks, each student can be represented as a point. Students with similar characteristics may appear close to each other.

Two important ideas: Linear models use lines or planes to divide feature space; distance-based models use distance to determine similarity.

MEMORY: GEOMETRIC = POINTS + SPACE + DISTANCE

17. Probabilistic Models

Probabilistic models use probability to represent uncertainty. Real-world data is uncertain, so a model may estimate something like “Spam probability = 0.90” rather than claiming absolute certainty.

Features and target variables can be treated as random variables. Probabilistic models can be considered in predictive and generative forms.

Predictive probabilistic model: Uses $P(Y|X)$, asking: Given X , what is the probability of Y ?

Generative probabilistic model: Models the joint distribution $P(X,Y)$, from which conditional or marginal probabilities can be derived.

Example: Naïve Bayes is a probabilistic classifier. Detailed classification algorithms belong to later units and are not expanded here.

MEMORY: PROBABILISTIC = PROBABILITY + UNCERTAINTY

18. Logical Models

Logical models use logical expressions or rules to divide data into groups. The reference material describes rule models and tree models.

Rule Model: Uses IF-THEN rules. Example: IF income is high AND credit history is good THEN loan approval is likely.

Tree Model: Rules are arranged in a tree structure. The tree asks questions step by step and reaches a final decision.

Models of Machine Learning



MEMORY: LOGICAL MODEL = IF → THEN

19. Grouping Models

Grouping means dividing data instances into groups according to similarity or common characteristics.

Example: Customers A and B are high spenders while C and D are low spenders. They can be divided into Group 1: High-spending customers and Group 2: Low-spending customers.

Purpose: Find similar instances, discover hidden patterns, organize data and understand data structure.

MEMORY: GROUPING = Which similar group?

20. Grading Models

Grading is different from grouping. Instead of asking “Which group does this belong to?”, grading asks “What score, rank or level should this receive?”

Example: Student performance can be represented as Excellent, Good, Average and Poor, or through numerical scores such as 90/100, 75/100 and 62/100.

Difference: Grouping places similar instances into groups; grading assigns a score, rank or level.

21. Parametric Models

A parametric model assumes a particular form of relationship and represents it using a fixed set/number of parameters.

Example: $Y = \beta_0 + \beta_1 X$, where β_0 is the intercept and β_1 is the coefficient/slope.

- Fixed form of model
- Uses parameters to describe the relationship
- Usually simpler
- Generally computationally efficient
- Example: Linear Regression (detailed regression is Unit II)

MEMORY: PARAMETRIC → PARAMETERS ARE FIXED

22. Non-Parametric Models

A non-parametric model makes fewer assumptions about the exact mathematical form of the relationship. It can be more flexible and adapt to the structure of data.

- More flexible
- Fewer assumptions about model shape
- Can model complex relationships
- May require more data
- Can be computationally more demanding

Examples include KNN and Decision Tree. Detailed classification material is outside Unit I.

Parametric	Non-Parametric
Assumes a fixed model form.	More flexible model form.
Fixed number of parameters.	Complexity can depend on data.
More assumptions.	Fewer assumptions.
Usually simpler.	Usually more flexible.
Often faster.	Can be computationally heavier.
Example: Linear Regression.	Example: KNN / Decision Tree.

MEMORY: PARAMETRIC → PARAMETERS ARE FIXED | NON-PARAMETRIC → MORE FLEXIBLE

23. Feature Transformation

Feature transformation means converting the original representation of features into another representation that is more useful for analysis or Machine Learning. One important feature transformation approach in your syllabus is dimensionality reduction.

24. What is a Feature?

A feature is an individual measurable property or characteristic of an observation.

Age	Attendance	Study Hours	Internal Marks
21	85%	4	25
22	90%	6	29

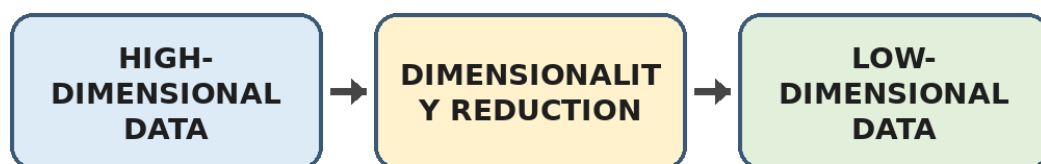
The features in this example are Age, Attendance, Study Hours and Internal Marks. Therefore, the dataset has 4 features/dimensions.

25. Dimensionality Reduction

If a dataset has Age, Salary and Experience, it has 3 dimensions. If it has 100 features, it has 100 dimensions.

Definition: Dimensionality reduction is the process of reducing the number of features/dimensions in a dataset while preserving as much important information as possible.

Dimensionality Reduction



Example: 100 features → dimensionality reduction → 10 important features/components.

26. Why do we need Dimensionality Reduction?

Reduce computational complexity: Fewer features generally mean less computation.

Reduce storage requirements: A smaller representation can require less storage.

Easier visualization: 50 or 100 dimensions are difficult to visualize, whereas 2 or 3 dimensions can be visualized more easily.

Remove redundancy: Some features may contain similar information, such as height in cm and height in meters.

Improve model efficiency: Reducing unnecessary/redundant information can make subsequent analysis or learning easier.

Reduce noise: Some dimensions may contain irrelevant or noisy information.

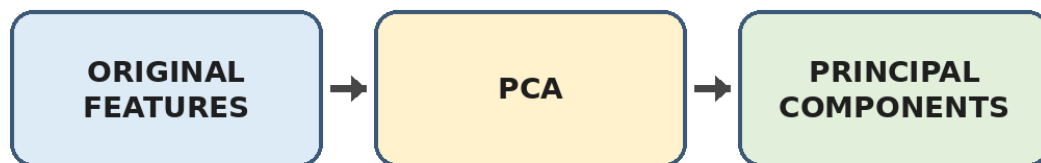
27. PCA — Principal Component Analysis

Full form: PCA = Principal Component Analysis.

PCA is a dimensionality reduction technique in your Unit I syllabus. Suppose you have many features X_1, X_2, X_3, X_4, X_5 . Some may contain related or overlapping information. PCA transforms the original features into new variables called Principal Components.

PCA can be understood as an orthogonal transformation that converts potentially correlated variables into linearly uncorrelated principal components.

PCA Concept



Most important concept: PC1 captures maximum variance; PC2 captures the next highest variance, and so on. The number of principal components is less than or equal to the number of original variables.

28. What is Variance?

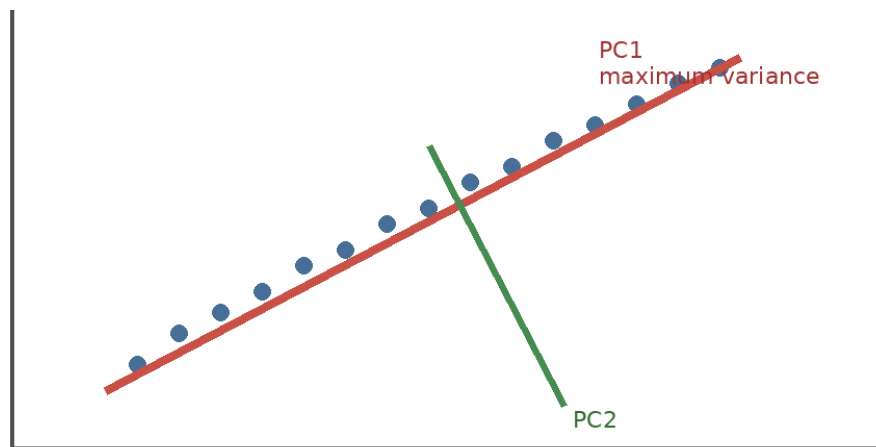
Variance tells us roughly how much the data spreads or varies. If data values are very similar, variation is low. If data values are widely spread, variation is high. PCA tries to find directions in which the data varies the most.

PCA tries to find directions in which the data varies the most.

29. PCA Visual Understanding

Imagine data points scattered in a feature space. PCA tries to find the most informative directions. PC1 is the direction of maximum variance and PC2 is the next important direction.

PCA: Principal Component Directions



30. Goals of PCA

- Identify patterns in data.
- Detect correlation between variables.
- Reduce dimensionality.
- Retain as much important variation/information as possible.

31. Steps of PCA

Step 1: Collect the data — Start with the original dataset.

Step 2: Standardize the data — Different features may have different scales. For example, Age may range from 18–60 while Salary may range from ₹20,000–₹2,00,000. Standardization is generally performed so scale differences do not dominate the result.

Step 3: Find relationships/variation — Examine how features vary together; this can be represented through covariance/correlation information.

Step 4: Calculate eigenvalues and eigenvectors — These help identify important directions/components. Eigenvector = direction; eigenvalue = importance/amount of variance in that direction.

Step 5: Sort eigenvalues — Arrange eigenvalues from highest to lowest and arrange corresponding eigenvectors in the same order.

Step 6: Select important components — Choose the required number k of principal components, for example 10 original features → 2 principal components.

Step 7: Construct projection matrix — Selected eigenvectors form the basis for projecting the original data.

Step 8: Transform the dataset — Project the original dataset into the selected lower-dimensional space.

PCA MEMORY CHART: $S \rightarrow E \rightarrow S \rightarrow T$ = Standardize → Eigenvalues/Eigenvectors → Select important components → Transform data.

32. Advantages of PCA

- Reduces dimensionality.
- Reduces redundancy.

- Makes visualization easier.
- Can improve computational efficiency.
- Can extract important information.
- Useful for high-dimensional datasets.

33. Limitation of PCA

Important limitation: PCA results depend on the scaling of variables. Therefore, scaling/standardization is important.

Another limitation is that principal components may not always be easy to interpret because they are combinations of original features.

34. Applications of PCA

- Data analysis
- Neuroscience
- Financial applications
- Feature extraction
- Visualization of high-dimensional data

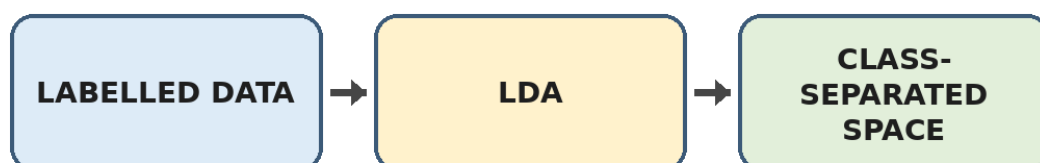
35. LDA — Linear Discriminant Analysis

Full form: LDA = Linear Discriminant Analysis.

LDA is another dimensionality reduction technique explicitly mentioned in your Unit I syllabus.

Basic idea: PCA asks “Which directions contain maximum variance?” LDA asks “Which directions separate different classes best?” This is the most important difference to remember.

LDA Concept



Example: Suppose students belong to Class A = Pass and Class B = Fail, with features such as Study Hours, Attendance and Internal Marks. LDA transforms the data so that Class A points stay together, Class B points stay together, and the two classes become as separated as possible.

36. Why is LDA called Supervised?

LDA requires class labels. For example, a dataset may contain Study Hours, Attendance and Class where each observation is labelled Pass or Fail. Because LDA knows the class of each observation, it can search for directions that separate those classes.

Study Hours	Attendance	Class
2	60%	Fail

Study Hours	Attendance	Class
3	65%	Fail
7	85%	Pass
8	90%	Pass

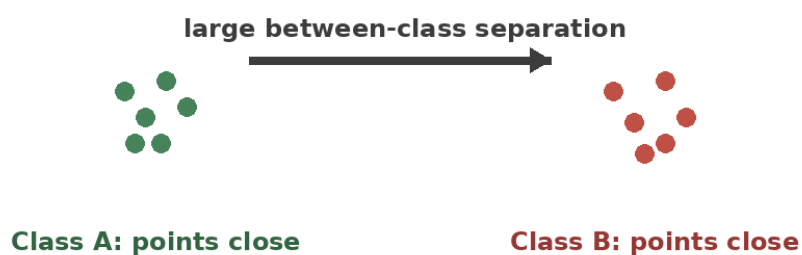
Definition: Linear Discriminant Analysis is a supervised dimensionality reduction technique that finds a lower-dimensional representation in which different classes are separated as effectively as possible.

37. Main Objective of LDA

LDA tries to maximize between-class separation and minimize within-class variation.

In simple words: different classes should be far apart, while members of the same class should remain close together.

! Within-Class Compactness and Between-Class Separat



38. Scatter Matrices in LDA

Within-Class Scatter: Measures how much observations vary inside the same class. Goal: minimize within-class scatter.

Between-Class Scatter: Measures how far apart different class means/groups are. Goal: maximize between-class scatter.

MEMORY: LDA = Classes close within themselves + Classes far from each other.

39. Basic Steps of LDA

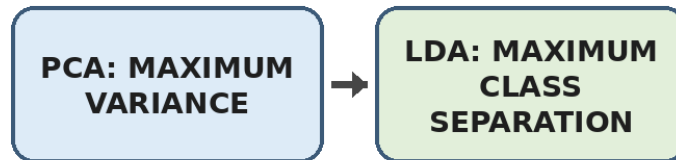
- Step 1: Collect labelled dataset.
- Step 2: Calculate the mean of each class.
- Step 3: Calculate within-class variation.
- Step 4: Calculate between-class variation.
- Step 5: Find directions that provide good class separation.
- Step 6: Select the required discriminant components.

Step 7: Project the original data into the lower-dimensional feature space.

Final result: High-dimensional labelled data → LDA → Lower-dimensional class-separated representation.

40. PCA vs LDA

PCA vs LDA — Main Idea

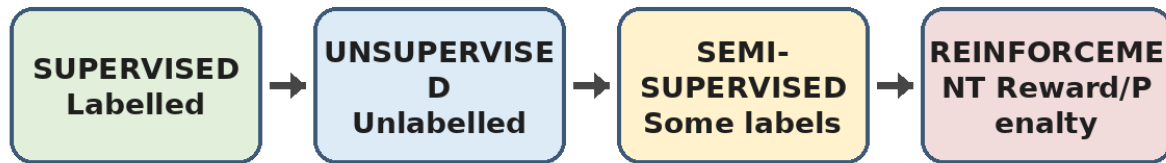


Point	PCA	LDA
Full form	Principal Component Analysis	Linear Discriminant Analysis
Type	Dimensionality reduction	Dimensionality reduction
Learning	Unsupervised	Supervised
Class labels	Not required	Required
Main objective	Maximum variance	Maximum class separation
Uses class information?	No	Yes
Focus	Overall data variation	Difference between classes
Within-class scatter	Not the main objective	Minimized
Between-class scatter	Not the main objective	Maximized
Application	Feature extraction, visualization	Classification preprocessing, feature extraction

GOLDEN LINE: PCA preserves maximum variance, whereas LDA maximizes class separability.

41. Unit I Complete Concept Map

Learning Paradigms



Machine Learning → Introduction → ML vs Traditional Programming → ML vs AI vs Data Science → Learning Tasks (Descriptive + Predictive) → Learning Paradigms (Supervised + Unsupervised + Semi-supervised + Reinforcement) → Models (Geometric + Probabilistic + Logical + Grouping/Grading + Parametric/Non-parametric) → Feature Transformation → Dimensionality Reduction → PCA + LDA.

42. Unit I — Exam Priority Chart

MUST PREPARE — TOP PRIORITY

1. Define Machine Learning and explain real-life applications.
2. Compare Machine Learning with Traditional Programming.
3. Differentiate AI, ML and Data Science.
4. Explain Supervised Learning with example.
5. Explain Unsupervised Learning with example.
6. Explain Semi-supervised Learning.
7. Explain Reinforcement Learning with example.
8. Compare supervised, unsupervised, semi-supervised and reinforcement learning.
9. Explain different models of Machine Learning.
10. Explain Geometric Models.
11. Explain Probabilistic Models.
12. Explain Logical Models.
13. Explain Grouping and Grading Models.
14. Differentiate Parametric and Non-parametric Models.
15. What is dimensionality reduction? Explain its need.
16. Explain PCA with steps.
17. Explain LDA with steps.
18. Differentiate PCA and LDA.

SECOND PRIORITY — Do not skip these smaller syllabus points: Descriptive tasks, Predictive tasks, Geometric model, predictive vs generative probabilistic idea, rule and tree logical models, grouping, grading, PCA goals, PCA limitation, LDA within-class scatter and LDA between-class scatter.

43. Last-Minute Memory Table

Topic	One-line memory
ML	Learn from data
Traditional Programming	Rules given by programmer
AI	Intelligent behavior
Data Science	Extract knowledge from data
Descriptive	Describe patterns
Predictive	Predict outcome
Supervised	Labelled data
Unsupervised	Unlabelled data
Semi-supervised	Some labelled + many unlabelled
Reinforcement	Reward/penalty
Geometric	Points, lines, distance
Probabilistic	Probability + uncertainty
Logical	IF-THEN rules
Grouping	Similar groups
Grading	Score/rank/level
Parametric	Fixed model form/parameters
Non-parametric	Flexible model
Dimensionality Reduction	Fewer features, important information retained
PCA	Maximum variance
LDA	Maximum class separation

44. How to Write SPPU 5/6-Mark Answers

For your 30-mark In-Sem, do not write one huge paragraph. Use a structured answer.

- Definition — 2-3 lines.
- Need / purpose — 4-5 points.
- Basic concept — explain the main idea.
- Diagram — draw/include the appropriate diagram.
- Steps — write the process in numbered points.
- Advantages — 4-5 points where applicable.
- Limitations — 1-2 points where applicable.
- Application/example — 1-2 examples.

Example: Explain PCA. Definition → Need of PCA → Basic concept → Diagram → Steps → Advantages → Limitation → Application/example.

45. Out-of-Syllabus Material Intentionally Not Expanded

The reference materials contain additional Machine Learning concepts, but these notes deliberately do not expand topics that belong outside your Unit I syllabus. Examples include Gradient Descent, Ordinary Least Squares, Logistic Regression details, Naïve Bayes details, KNN details, SVM details, Random Forest and Boosting. Regression is Unit II and classification/clustering/ensemble topics belong to later

units. This keeps the notes focused on your in-sem pattern.

Likewise, the PCA/LDA reference material is used to explain standardization, eigenvalues/eigenvectors, principal components, within-class scatter and between-class scatter without adding advanced mathematics that is not necessary for this Unit I preparation.

46. Unit I Completion Check

Official syllabus point	Status
What is Machine Learning	Covered
Definitions	Covered
Real-life applications	Covered
ML vs Traditional Programming	Covered
ML vs AI vs Data Science	Covered
Descriptive Tasks	Covered
Predictive Tasks	Covered
Supervised Learning	Covered
Unsupervised Learning	Covered
Semi-supervised Learning	Covered
Reinforcement Learning	Covered
Geometric Model	Covered
Probabilistic Models	Covered
Logical Models	Covered
Grouping Models	Covered
Grading Models	Covered
Parametric Models	Covered
Non-parametric Models	Covered
Dimensionality Reduction	Covered
PCA	Covered
LDA	Covered
Important questions	Included
Memory charts	Included
Diagrams	Included

47. Ten Things to Remember Before the Exam

1. ML = Learning patterns from data.
2. Traditional programming = Rules given.
3. Supervised = Labels available.
4. Unsupervised = Labels unavailable.

5. Semi-supervised = Some labels + many unlabelled examples.
6. Reinforcement = Reward/Penalty.
7. Geometric = Points/Distance.
8. Probabilistic = Probability/Uncertainty.
9. Logical = IF-THEN.
10. PCA = Maximum Variance | LDA = Maximum Class Separation.

UNIT I COMPLETE — aligned to the syllabus shared for SPPU Fourth Year AIDS, 2020 Course, 417521 Machine Learning.